# Evidence Lab Teaching Guide - Version 1.0

*For librarians, faculty, instructors, and instructional designers - spoiler-free*

## 1. What Evidence Lab is

Evidence Lab is an institution-neutral information-literacy investigation game for college students. Students encounter realistic information problems, choose what evidence to inspect, make a decision, explain which evidence mattered, reflect on confidence, and receive feedback.

## 2. Ways to use Evidence Lab

- 2-minute challenge - one quick investigation.
- 5-minute challenge - two investigations.
- 10-minute challenge - about four investigations.
- 20-minute challenge - about seven investigations.
- Choose individual cases - instructors or students can select any of the 36 cases.
- Challenge times are estimates, not countdowns; students are not penalized for taking longer.

## 3. Four case levels

- Guided (Cases 1-4): verification and uncertainty calibration.
- Networked (Cases 5-12): source tracing and information chains.
- Contextual (Cases 13-28): provenance, authority, currency, stakes, visual evidence, health, careers, and institutional information.
- Synthesis (Cases 29-36): data, breaking news, source independence, translation, AI citation tracing, and bounded synthesis.

## 4. Learning outcomes

1. Verify provenance and existence.
2. Select evidence strategically.
3. Evaluate claim-evidence fit.
4. Trace information networks and source dependence.
5. Evaluate authority contextually.
6. Interpret visual, quantitative, and multimodal evidence.
7. Calibrate uncertainty.
8. Update conclusions responsibly.
9. Make consequential information decisions.
10. Communicate bounded conclusions.

## 5. Scoring

Each investigation is worth 10 Investigator Points: Decision 4, Evidence 3, Reasoning 2, Investigation Strategy 1. Confidence is recorded but not scored.

- Decision: Does the conclusion fit what the evidence warrants?
- Evidence: Did the student investigate evidence capable of resolving the question?
- Reasoning: Did the evidence the student selected as important genuinely support the decision? Students may select multiple pieces.
- Investigation Strategy: Did the investigation include high-value evidence capable of resolving the question? Responsible additional verification is not penalized.
- Confidence: A reflection/calibration measure, not a reward for certainty.

## 6. Investigation Log

Completed attempts from challenges and individually selected cases are stored in the browser using local storage. No sign-in is required. The log persists when the student closes and reopens the site on the same browser/device, unless browser/site data are cleared. Repeat attempts are retained rather than overwritten.

- Students can Print / Save PDF, Download CSV, or Copy Summary.
- The log includes decision, evidence opened, evidence selected as important, confidence, score breakdown, consequence, and transfer principle.
- Evidence Lab does not automatically transmit the log to the instructor, institution, or creator.
- For assignments, students can export the log and submit it through the institution's existing LMS.
- Because the record is browser-local and modifiable by a technically knowledgeable user, use it for formative/low-stakes work rather than secure high-stakes testing.

## 7. Suggested teaching patterns

- Warm-up: 2-minute challenge followed by one discussion question.
- Library session: 10-minute challenge plus debrief on evidence selection.
- Course assignment: instructor-selected cases plus Investigation Log submission.
- Extended workshop: 20-minute challenge followed by small-group comparison of reasoning paths.
- Asynchronous: students complete work over several visits on the same browser and submit the log.

## 8. Choosing cases

Use the Cases page to search, filter by level, jump directly to a case number, and see which cases have been completed in the current browser. The page remembers the student's approximate scroll position when they return from a case.

## 9. Alignment with the ACRL Framework (2026 revision)

Evidence Lab is informed by and aligned with the ACRL Framework for Information Literacy for Higher Education, including revisions approved September 2, 2026. The game is not intended as a comprehensive implementation or assessment of every Framework knowledge practice or disposition. Coverage is intentionally uneven because cases are designed around authentic information problems rather than a six-frame quota.

- Authority Is Constructed and Contextual — very strong: students evaluate expertise, credibility, incentives, institutional authority, and AI authority in context.
- Information Creation as a Process — very strong: students trace how posts, images, clips, translations, summaries, citations, and other information products are created, transformed, and circulated.
- Information Has Value — selective but meaningful: sponsorship, financial incentives, access barriers, money, personal information, attribution, and information exchange appear in relevant cases.
- Research as Inquiry — very strong: students investigate uncertainty, identify gaps, gather evidence, update conclusions, and sometimes conclude that available evidence is insufficient.
- Scholarship as Conversation — strong: students trace claims and citations, distinguish independent corroboration from repetition, and compare research findings with later representations.
- Searching as Strategic Exploration — very strong: students choose what to check, trace sources laterally, use authoritative channels, revise search paths, and decide when evidence is sufficient.

The 2026 revision is especially relevant to Evidence Lab because it explicitly addresses emerging technologies and AI, metacognition, evolving information creation processes, and critical evaluation of AI-mediated searching and authority.

Framework reference: Association of College and Research Libraries. Framework for Information Literacy for Higher Education. Adopted 2016; revisions approved September 2, 2026.

## 10. Accessibility and multimodal evidence

Evidence Lab uses text alternatives and accessible descriptions for essential visual information. Multimodal cases use progressive disclosure: students initially see the encountered artifact and reveal additional evidence only when they choose the corresponding investigation action.

## 11. Privacy and assessment

The public GitHub Pages edition requires no student account and does not automatically collect or transmit student activity. Browser-local persistence reduces data collection but means logs do not automatically follow students to another device. Students should export a copy when they need to preserve or submit work.

## 12. Reuse and localization

The educational content and teaching materials are licensed CC BY 4.0; original software/code is licensed MIT. Evidence Lab is designed to work without institution-specific systems. Local adaptations should preserve the evidence logic, accessibility, and spoiler separation.

## 13. Spoiler-free 36-case catalog

| Case | Title | Level | Environment |
|---|---|---|---|
| 1 | Just Put It in the Bibliography | Guided | 💬 Group Project Chat |
| 2 | Real Article, Bigger Claim | Guided | 💬 Group Project Chat |
| 3 | ChatGPT Is Always Wrong | Guided | 📚 Study Session |
| 4 | But It Says 100% | Guided | 🤖 AI Conversation |
| 5 | I Didn't Read It, But... | Networked | 💬 Before Class |
| 6 | GUYS DID YOU HEAR?? | Networked | 📱 Group Chat |
| 7 | Everybody Is Saying It | Networked | 📡 Across Your Feeds |
| 8 | The Rumor Was True | Networked | 💬 Student Chat |
| 9 | You Already Shared It | Networked | 🔔 Social Update |
| 10 | I Found the Perfect Article...But I Can't Read It | Networked | 📚 Library Research |
| 11 | The Expert Is Outside Their Expertise | Networked | 📱 Social Post |
| 12 | DON'T STUDY AFTER 10 PM | Networked | 📱 Study Group |
| 13 | Wait...Is That Here? | Contextual | 📸 Social Image |
| 14 | The Cropped Protest Photo | Contextual | 📰 Image + News |
| 15 | Is This AI-Generated? | Contextual | 🖼️ Synthetic Media |
| 16 | The Video Starts 12 Seconds Late | Contextual | 🎥 Viral Video |
| 17 | SCIENCE PROVES MUSIC HELPS YOU STUDY | Contextual | 📱 Social + Research |
| 18 | My Favorite Creator Says This Supplement Works | Contextual | 📱 Influencer + Health |
| 19 | This Creator Actually Knows Their Stuff | Contextual | 📱 Social Video |
| 20 | The Headline Says Coffee Is Good Again | Contextual | 📰 News + Health |
| 21 | Congratulations! You Won a Scholarship | Contextual | 📧 Scholarship Email |

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|---|---|---|---|
| 22 | Your Account Will Be Locked Today | Contextual | 📧 University-Looking Email |
| 23 | A Recruiter Just DM'd Me | Contextual | 💼 Recruiter DM |
| 24 | $35 an Hour. Remote. No Experience. | Contextual | 📱 Job Text |
| 25 | The Internship Is Real…But Is This Listing? | Contextual | 💼 Job Board |
| 26 | Professor Moved the Exam Online | Contextual | 💬 Course Chat |
| 27 | CAMPUS CLOSED TOMORROW | Contextual | 📱 Screenshot + Alert |
| 28 | Official university Page Says It's Free | Contextual | 🔎 Web Search |
| 29 | This Chart Looks Terrifying | Synthesis | 📊 Social Chart |
| 30 | 90% of Students Agree! | Synthesis | 📊 Poll + News |
| 31 | BREAKING: Nobody Knows Yet | Synthesis | 📰 Breaking News |
| 32 | Five News Sites Can't All Be Wrong | Synthesis | 📰 News Ecosystem |
| 33 | Mom Sent This in the Family Chat | Synthesis | 👨‍👩‍👧 Multilingual Family Chat |
| 34 | The Quote Is Real—But What Came Before? | Synthesis | 🌐 International News |
| 35 | AI Search Quoted the Article | Synthesis | 🤖 AI Search |
| 36 | AI Tutor Beat the Professor! | Synthesis | 🤖 AI + Research + Social |

## 14. Public vs. confidential educator materials

This Teaching Guide is intentionally spoiler-free and may be shared publicly. The separate Evidence Lab Instructor Key contains case resolutions and protected teaching metadata and should not be placed in the public repository, public LibGuide, or Zenodo release.

## 15. Citation

Liu, Y. (2026). Evidence Lab: Follow the evidence. Make the call (Version 1.0) [Educational software]. https://yanlingliuinfolit.github.io/evidence-lab/

A DOI citation will be added after the Version 1.0 Zenodo archival release.